

Retailligence

Forecasting next week's sales for every department of every Walmart store, from sales history, store, product, promotion, month and holiday data, to plan stock and staff a week ahead.

421,570

sales rows

45 × 81

stores × departments

\$1,277

average miss / dept-week

0.987

test R^2 (share of the ups and downs the model explains; 1 = perfect)

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Problem, data and what we predict

Group 12 – Retail Sales Prediction. Build a model that predicts retail sales using historical sales, product, store, promotion, month, holiday and other suitable information.

Target *(the number we predict):* **next week's sales**
`Weekly_Sales` of one department in one store, in dollars:
supervised *(learns from examples with known answers)* **regression**
(predicting a number, not a category) on time-ordered data, so we never train on the future.

Data: Walmart, Kaggle *(data-science competition website)*
(2014)
`train.csv` + `features.csv` + `stores.csv` , joined on store and week.

421,570

sales rows

3,331

store-department series

143

weeks · 2010-02 → 2012-10

45

stores · types A/B/C

81

departments (products)

\$6.74B

total sales

DATA TYPE	COLUMNS
Categorical (ID)	Store
Categorical (nominal)	Dept , Type
Date/time	Date
Numerical, continuous	Weekly_Sales , MarkDown1-5 , Temperature , Fuel_Price , CPI , Unemployment
Binary	IsHoliday
Numerical, discrete	Size

Cleaning, preprocessing and features

Cleaning

1,285 negative sales (returns) → 0 · missing markdowns (*price cuts / promotions*) → 0 + a "reported" flag · weeks with no record count as 0 sales in the history · no duplicates.

History features *(input columns the model uses)*

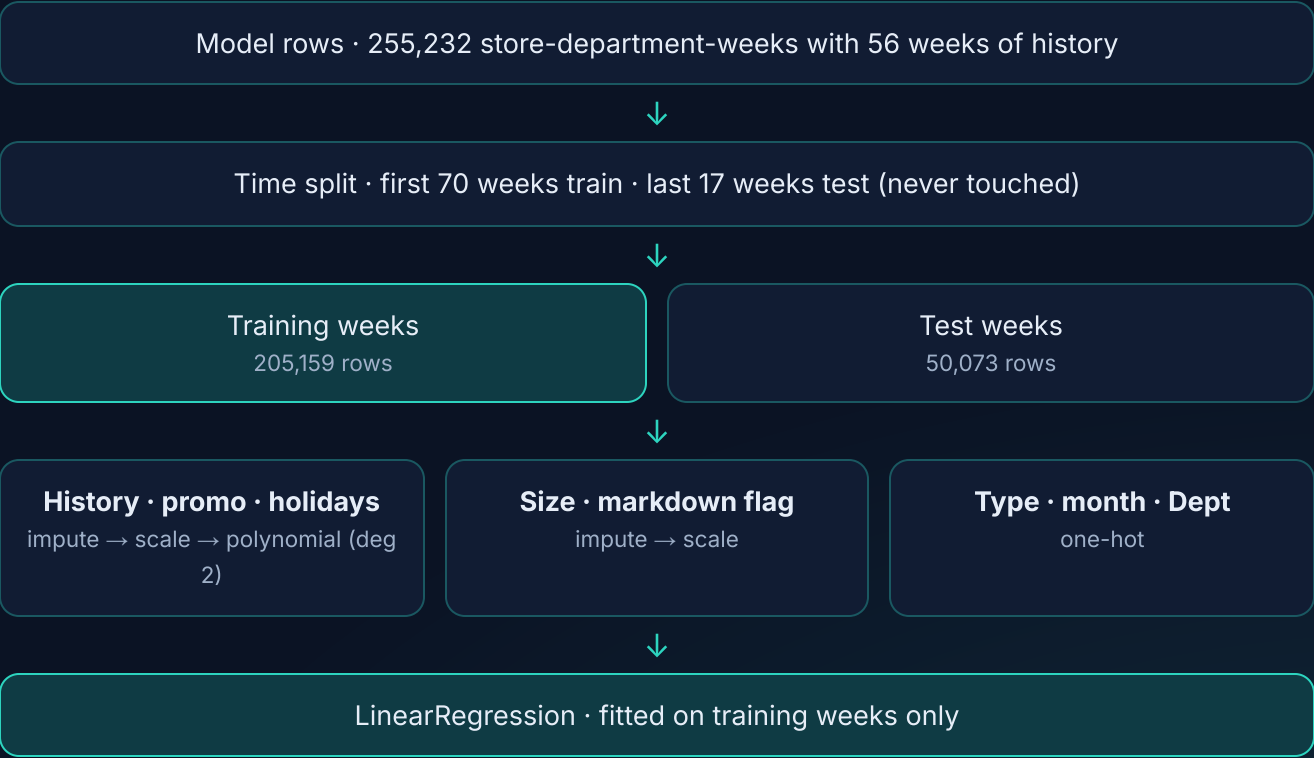
last week, 2 weeks ago, 4-week average, same week last year, and *last year's jump* for this week.

Date/time extraction

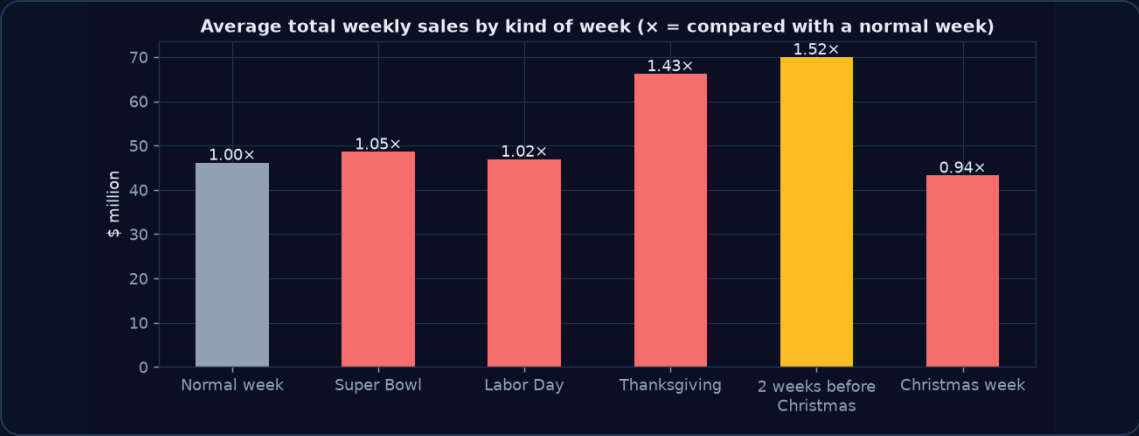
month, 4 holiday flags and the 2 pre-Christmas weeks (the biggest weeks, not flagged in the data).

Leakage-free *(no future information sneaks into training)* pipeline *(the same steps, in the same order, for every row)*

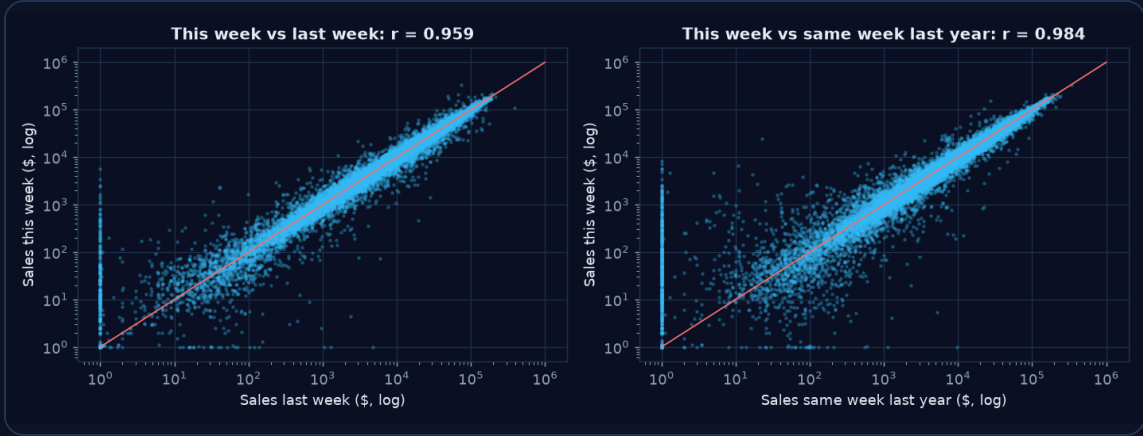
impute (*fill in missing values*) → scale (*put all numbers on the same scale*) → polynomial terms, one-hot (*one yes/no column per category*) for type, month and department; fitted on training weeks only.



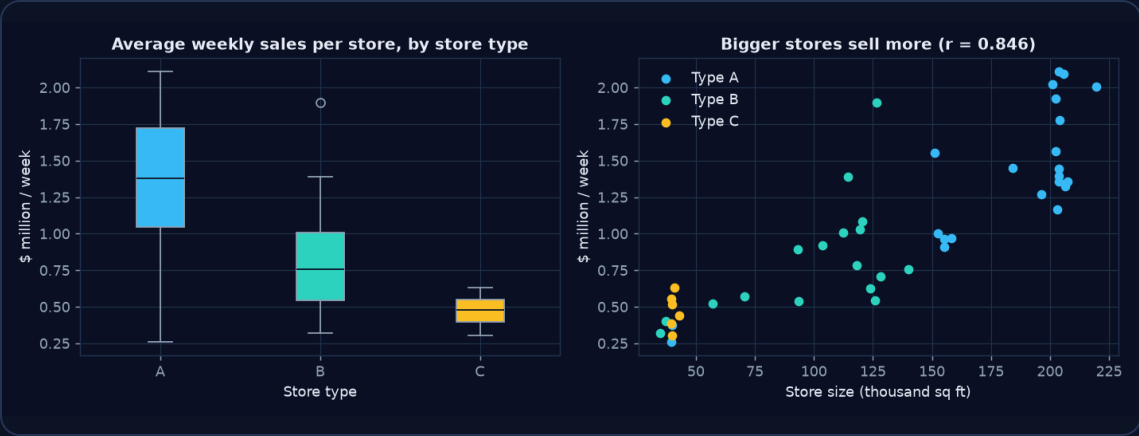
What the data tells us



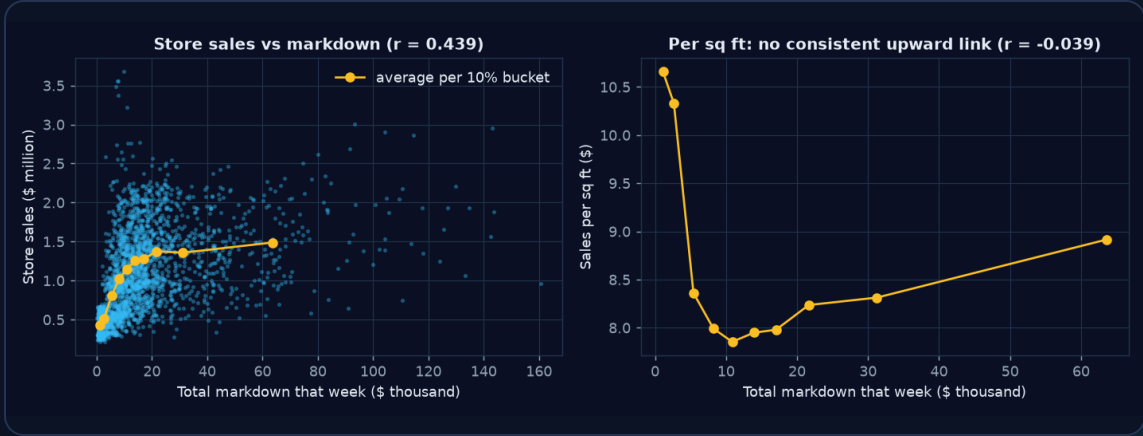
Holidays multiply sales: Thanksgiving 1.43x, pre-Christmas 1.52x a normal week.



History is the strongest clue: $r =$ (correlation, -1 to 1) 0.959 with last week, 0.984 with the same week last year.



Store and product matter: Type A sells most; top 10 departments = 46.0% of sales.



Promotions look strong, but it's store size: per sq ft $r = -0.039$.

Validation design and models

Never peek at the future

The last 17 weeks are locked away as the test set; 5 expanding-window (each check trains on all earlier weeks) folds inside the training weeks always learn from the past.

Three rule-of-thumb baselines (simple rules the model must beat)

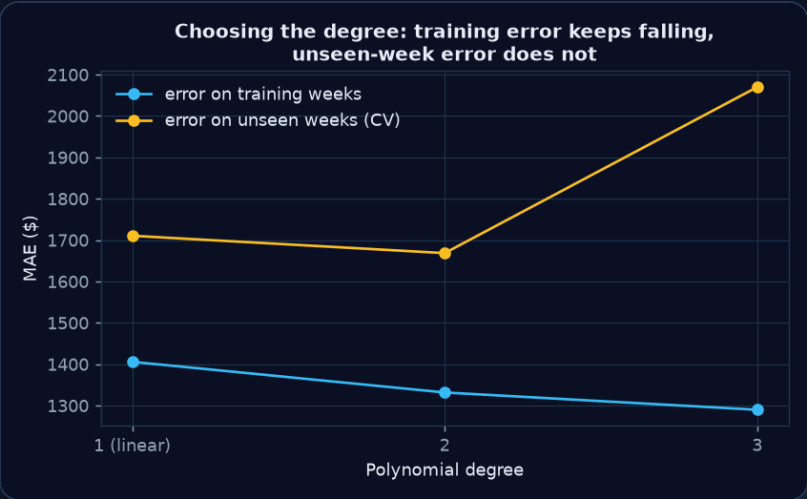
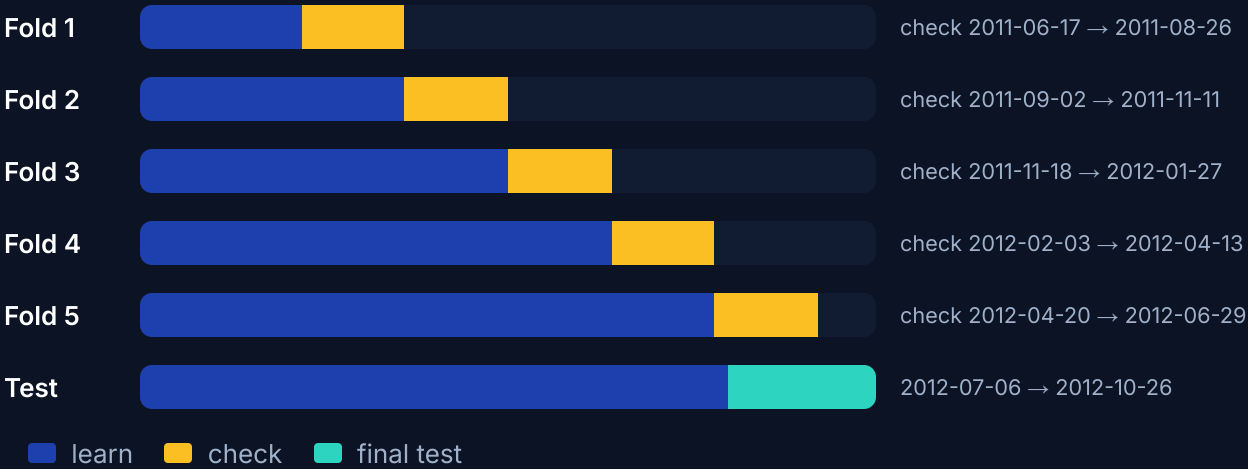
last week · same week last year · 4-week average + last year's jump.

Linear vs Polynomial Regression (linear regression plus squared and multiplied inputs, so the line can bend)

degree-2 terms like *last week* × *Thanksgiving* let a holiday multiply sales.

Degree 2 wins

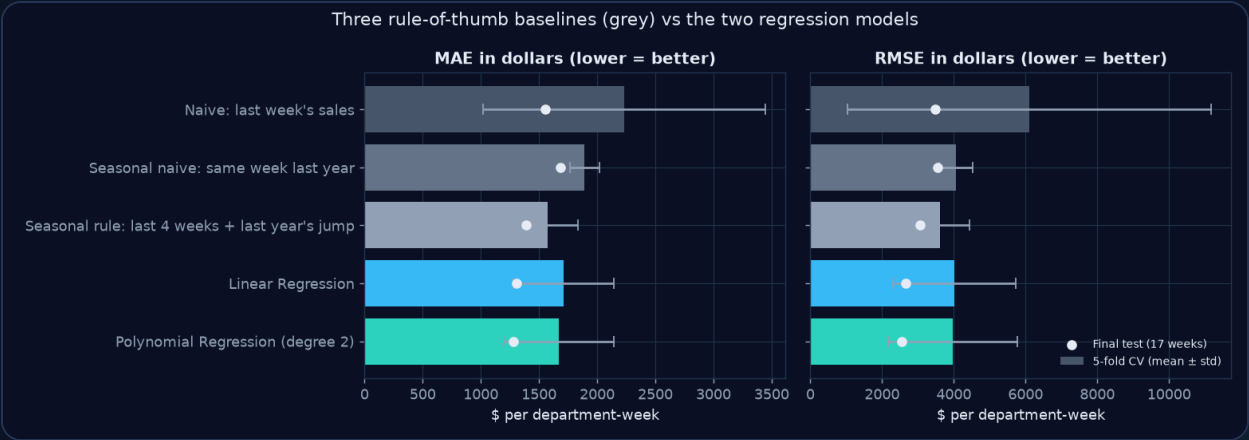
CV (cross-validation: testing on held-back blocks of weeks) MAE (average miss in dollars): degree 1 \$1,710 · **2 \$1,668** · 3 \$2,070 (over-fits (memorises training weeks, does worse on new weeks)).



Results on 17 unseen weeks

METHOD	CV MAE	TEST MAE	TEST RMSE	TEST MSE	TEST R ²	ERROR CUT VS NAIVE
Naive: last week's sales	\$2,232	\$1,555	\$3,490	12.2M	0.975	+0.0%
Seasonal naive: same week last year	\$1,893	\$1,683	\$3,550	12.6M	0.974	-8.2%
Seasonal rule: last 4 weeks + last year's jump	\$1,575	\$1,386	\$3,062	9.4M	0.981	+10.9%
Linear Regression <i>(a weighted sum of the inputs)</i>	\$1,710	\$1,309	\$2,655	7.1M	0.985	+15.8%
Polynomial Regression <i>(linear regression plus squared and multiplid inputs, so the line can bend)</i> (degree 2)	\$1,668	\$1,277	\$2,544	6.5M	0.987	+17.9%

Key: MAE = average miss in dollars · RMSE = typical miss, big misses punished more · MSE = average of the squared misses · R² = share of the ups and downs explained (1 = perfect) · CV = cross-validation, testing on held-back blocks of weeks · naive = next week equals last week



Reading the results honestly

The model wins on the test weeks

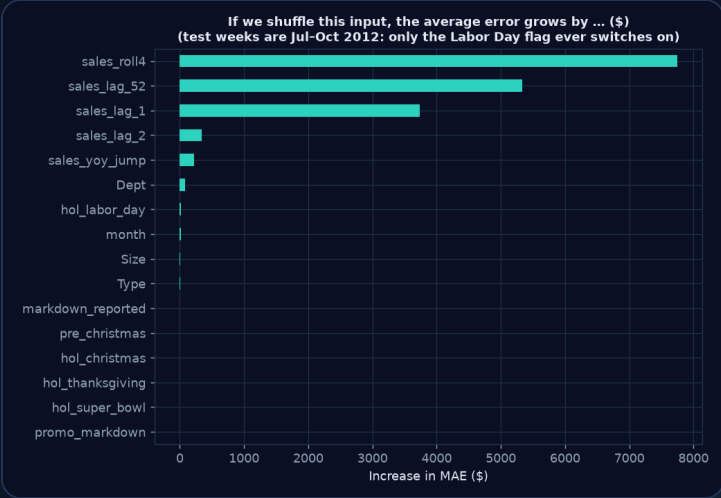
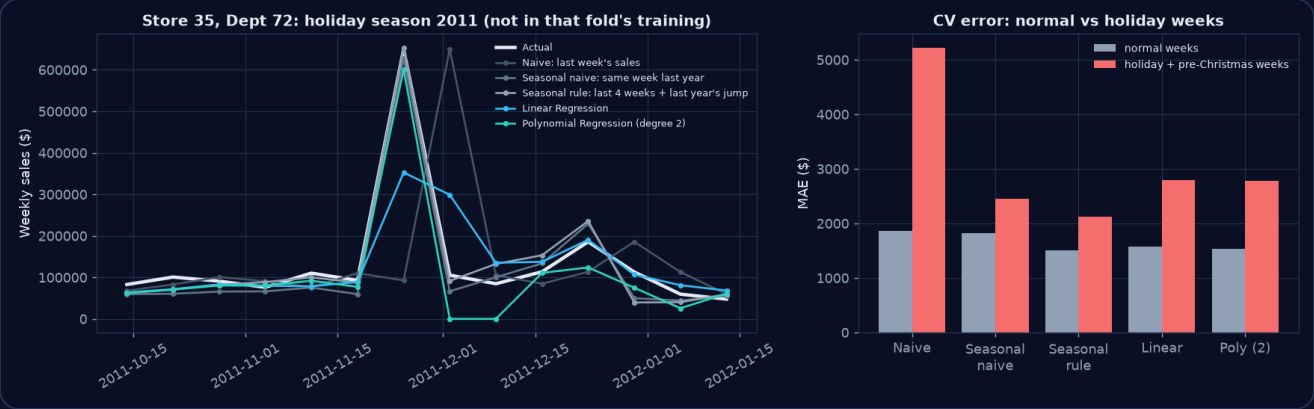
\$1,277 vs \$1,386 for the best rule: 7.9% lower error.

The rule wins big holiday weeks

Only one holiday season exists to learn from. Holiday-week CV (cross-validation: testing on held-back blocks of weeks) MAE (average miss in dollars): rule \$2,120, model \$2,785.

What drives the forecast

4-week average (+\$7,753 MAE if shuffled), same week last year (+\$5,332), last week (+\$3,733); markdowns (price cuts / promotions) ≈ nothing.



Streamlit app and testing

In: store, department, week

plus what-if markdown (*price cut / promotion*), kind of week and last week's sales, limited to the training range.

Out: forecast + 80% range (*8 out of 10 past forecasts landed inside it*)

with a rule-of-thumb cross-check, actual sales for past weeks, a 16-week chart and the model's test scores.

Honest by design

training weeks are labelled "not a fair test"; big holidays carry a warning; departments without 56 weeks of history get an error, not a number.

21 / 21 tests pass

feature logic, no leakage (*future information sneaking into training*), saved model, and the app driven with 5 different inputs.



Conclusion, limitations and future scope

Conclusion

Degree-2 Polynomial Regression (*linear regression plus squared and multiplied inputs, so the line can bend*) forecasts next week's department sales with R^2 (*share of the ups and downs the model explains; 1 = perfect*) 0.987 and an average miss of \$1,277, 17.9% better than "same as last week".

Limitations

one holiday season to learn from · one week ahead only · anonymised markdowns (*price cuts / promotions*) · polynomial terms over-react far outside the training range.

Future scope

more years of data · a holiday-week model · 2–4 weeks ahead · Ridge-regularised (*extreme weights are penalised to avoid over-fitting*) terms · live POS (*shop checkout data*) feed.



References: Fildes et al. (2022) IJF 38(4) · Makridakis et al. (2022) IJF 38(4) · Bojer & Meldgaard (2021) IJF 37(2) · Bergmeir & Benítez (2012) Inf. Sci. 191 · Hyndman & Athanasopoulos (2021) FPP3 · Kaggle (*data-science competition website*) Walmart Store Sales (2014).